Tapeout Guide

This document is a *work-in-progress* guide for taping out a simple RISC-V core from Chipyard with TSMC's 28nm process. It is intended to be a living document. Please correct errors and ambiguities, and leave comments for questions and improvements!

Machine/Software Access

These are the main resources you need access to for the tapeout tools:

1. ssh access to *acghaswell16* and *acgcascade52* servers. You must be on the Penn VPN to connect to these servers.
    a. You need to send Joe your SSH key to access these servers. See this guide for generating an SSH key.
    b. the specific command is **ssh PENNKEY@acghaswell16.seas.upenn.edu** where you replace PENNKEY with your actual PennKey (the part of your school email before the @ symbol)
2. Slack access: we use the riscv-tapeout channel of the Radway-Lab workspace
    a. Joe will add you
3. access to our private GitHub repo of rflowgen
    a. You need to send Joe your GitHub username
4. access to Cadence Support Portal
    a. If you can already login to https://support.cadence.com, you are all set!
        i. Check that you can see documentation for following products: Genus, Innovus and Conformal. If you can't, contact Joe to get your access upgraded.
    b. If you do not have an account, you can self-enroll in a Cadence ASK account for academia.
        i. Create your Cadence account and verifying your academic email address
        ii. Select "Academic or Content Only Access"
        iii. Enter your institution details to complete enrollment
    c. If you either do not see the "Academic or Content Only Access" or if you receive the "HostID expired" message, contact Joe to get your access reinstated.
5. sign TSMC Process Design Kit (PDK) Non-Disclosure Agreement (NDA)
    a. You can (and should) get started on the steps below without needed TSMC PDK access - you'll start with an open-source freepdk
    b. check on Slack to coordinate this with Prof. Radway

Unix Tips

For this project, you will often be connecting to remote Linux servers and running programs on those servers. There are many things to learn about the Linux command line, but using **tmux** is critical for productivity and to avoid losing your work if your WiFi or VPN flakes out. RedHat has a nice tmux tutorial. Note that tmux is already installed on all the machines you'll use.

You will also find it helpful to use a command-line text editor from time to time. **nano** is a basic one, but it would be a good idea to learn either **emacs** or **vim**, which are very fully-featured editors with good command-line interfaces.

On cascade52, you can use **modulefiles** to make various versions of the CAD tools available. Run **module avail** to see the list of available modules.

- **module load ic/latest** loads the Cadence IC package (which contains Virtuoso) at the latest version. Now you can run the **virtuoso** program.
- **module list** shows which modules you currently have loaded.
- **module unload ic** unloads the ic module, so you can then switch to another version (though we only have one version at the current time).

Graphical Desktop on cascade52

Using a graphical desktop environment is often helpful, as the CAD tools have helpful graphical interfaces (GUIs). First, install a VNC viewer to access the graphical Linux desktop running via the VNC server that is on acgcascade52. TigerVNC is a nice open-source VNC viewer for Mac, Windows and Linux. Somewhat confusingly, TigerVNC is also the name of a VNC *server* (from the same developers), but you will only need the *viewer* on your laptop.

- Mac/Linux
    - Setup an SSH tunnel from your laptop to acgcascade52. On your laptop, use the terminal to run the command **ssh -N -L /tmp/vnc-local.sock:/tmp/vncserver-PENNKEY.sock PENNKEY@acgcascade52.seas.upenn.edu**, replacing PENNKEY with your actual PennKey. This creates a tunnel from a socket on acgcascade52 to a local socket on your laptop at **/tmp/vnc-local.sock**
    - Connect your VNC client to **/tmp/vnc-local.sock** and you should see a Linux desktop appear.
    - You should be prompted to enter a password, which is BananaZoo6
- Windows
    - Setup an SSH tunnel from your laptop to acgcascade52. On your laptop, open the Command Prompt program and run the command **ssh -N -L**

5901:/tmp/vncserver-PENNKEY.sock PENNKEY@acgcascade52.seas.upenn.edu, replacing PENNKEY with your actual PennKey. This creates a tunnel from a socket on acgcascade52 to port 5901 on your laptop.
    - Connect your VNC client to **localhost:5901**. You should see a Linux desktop appear.
    - You should be prompted to enter a password, which is BananaZoo6

chipyard Elaboration and Simulation

We will use chipyard to generate the Verilog code for our chip: the RISC-V core, caches, memories and other peripherals. chipyard uses the Chisel language, an alternative to Verilog/SystemVerilog. The Chisel code for your chip has to be translated into Verilog so that other CAD tools can use it: this process is known as *elaboration*. chipyard installation takes about 2 hours to run.

chipyard is run on the **acghaswell16** machine (*not* acgcascade52). In your home directory on **acghaswell16** follow the chipyard setup instructions. You will need to install conda, and can then install chipyard. You can skip the steps relating to FireSim and FireMarshal as we won't use those components, by using this command:

```
./build-setup.sh --skip-firesim --skip-marshal
```

After you reach **Step 1.4.5. What's Next?** you should edit the file **variables.mk** (in your top-level **chipyard** directory) on line 96 and change **RocketConfig** to **TinyRocketConfig** to make the design smaller and easier to work with. Then proceed to the RTL Simulation guide and use the Verilator simulator (which is already installed) to run the **asm** tests (in parallel) with:

```
make
make run-asm-tests
```

Next, you will need to aggregate the many Verilog/SystemVerilog files that comprise the TinyRocket chip into a single **design.v** file, which is what rflowgen needs:

```
/chipyard/sims/verilator/generated-src/chipyard.harness.TestHarness.TinyRocketConfig/gen-collateral
```

```
cd generated-src/chipyard.harness.TestHarness.TinyRocketConfig/gen-collateral
echo `pwd`/chipyard.harness.TestHarness.TinyRocketConfig.top.mems.v >> ../chipyard.harness.TestHarness.TinyRocketConfig.top.f
python3 ~devietti/cat-files.py ../chipyard.harness.TestHarness.TinyRocketConfig.top.f > design.v
```

rflowgen Setup

The **acgcascade52** server is where you will run the rest of the tapeout flow (everything after Chipyard). In your home directory on **acgcascade52**, clone the private GitHub repo https://github.com/radway-lab/rflowgen. Once you start making edits to the repo, please work on a private branch (or GH fork) so that we do not trample on each other's work, but can also see what others have done to debug when (inevitably) something works for one person but not for another. We can also share code by merging commits from each others' branches, and via pull requests.

Next, you need to ensure that rflowgen is set to use the freepdk. In the file **src/myflows.rs**, ensure the function **rflowgen_current_flow()** looks like this:

```
pub fn rflowgen_current_flow() -> FlowGraph {
    freepdk_flow()
}
```

Once you have switched to your branch, follow the rflowgen setup instructions. Use the **design.v** that you got from Chipyard (above). For your **constraints.sdc** file, you can copy **/home/devietti/constraints.sdc** into your design directory.

Synthesis

To start running rflowgen, do the following:

```
module load genus
cd <your-build-directory>
rflowgen run --through cadence_genus_synthesis
```

Note: rflowgen puts logs into the directories for each step, e.g., the synthesis log file is in **<your-build-directory>/2_cadence_genus_synthesis/logs/genus.log** so you don't need to

scroll through your terminal to see what happened. It can be helpful to copy these logs elsewhere, too, to record previous error messages while you tinker with the build. Running **rflowgen clean cadence_genus_synthesis** will delete the specified step directory (like **2_cadence_genus_synthesis**) and the logs inside.

Synthesis should take about an hour to run.

Place and Route

To run place and route, first load Cadence Innovus, Tempus and Voltus onto your PATH, then run rflowgen:

```
module load innovus
module load ssv
rflowgen run --through cadence_innovus_signoff
```

PnR takes >6 hours to run.

A useful guide to some Stylus FlowKit internals is in Cadence's Innovus Foundation Flow to Stylus Flowkit Migration Guide.

Design Rule Checks (DRC) and Logic-vs-Schematic (LVS)

First, make the Mentor Calibre tool available by running:

```
module load calibre/2024.1_36
```

Then, run the DRC/LVS steps via rflowgen.

TODO: TSMC PDK Integration

Work-in-progress: connect rflowgen to TSMC 28nm
N40 full-custom example: https://github.com/radway-lab/HYDRA

Troubleshooting

Error: Conda environment directories already exist!

You may run into this issue if you end up interrupting the Chipyard setup. If you do, just remove the `.conda-lock-env` directory (`rm -rf .conda-lock-env` from the same directory you ran the setup in) then re-run the setup.

The setup will suggest you "... skip this step with -s 1". Don't do this because you'll skip creating a conda environment and grabbing the dependencies you need.

Missing <package> command during setup

If you get this during the Chipyard setup, you likely need to just run the setup process again.Follow the instructions from Error: Conda environment directories already exist!

Moved ssh key and can't clone from Github

If you move your .ssh key to the new server, Github will prevent transactions if the file permissions are wrong on the ssh key. You need to set the file permissions on the key on the remote using `chmod`:

```
$ chmod 600 ~/.ssh/<your key>
```

Error running `make <build-step>`

Try running `make clean-all`

Run SFK steps "by hand" for faster debugging iterations

Rerunning an entire rflowgen step can be painfully slow. E.g., I am currently hitting lots of silly errors in the postroute step, like missing flags, undefined variables, etc. Stylus FlowKit has much finer-grained steps, however, and saves a bunch of intermediate database checkpoints as well. I have found it useful to switch from rflowgen to flowtool to run just the SFK step that is crashing. For example, I can repro bugs in generate_signoff_results in about 80 seconds, while starting rflowgen's postroute from scratch would be well over an hour.

First, change to the step working directory of the rflowgen step that is failing. Say postroute isn't working; do the following:

```
cd a_cadence_innovus_stylus_postroute/innovus-stylus-build/
```

Next, get the names of the SFK steps that still need to be run. Usually the error step will be shown by running:

```
flowtool -files flow.yaml -status
```

Finally, run the failing SFK step. flowtool will automatically backtrack to its most recent db checkpoint and run forward from there (note that rflowgen sets a variety of environment variables that Innovus needs, so you may well need to export these by hand as with **design_name** below):

```
design_name=ChipTop flowtool -files flow.yaml -from MY_STEP -to MY_STEP
```

This process can be further accelerated by commenting out irrelevant steps in **flow.yaml**. All the files in the **innovus-stylus-build/** directory are copies of the original files in **src/recipes/cadence_innovus_flowsetup/**. Often when debugging we want to edit something and quickly try out the changes. Editing the originals in git won't affect the SFK working directory. You can re-run the rflowgen flowsetup step to create a new working copy based on the files in git, but this will force re-running of *all* subsequent steps, starting from floorplanning, taking hours of time to get back to postroute or signoff.

A better option is to edit the SFK working directory files directly. One must, however, remember to copy the (successful) changes back to git later. A nicer option is to temporarily hardlink the files from git to the SFK working directory - this way, edits in one place will automatically appear in the other as there is just one underlying file in both places. git will track all changes and they can easily be committed. The script **nodes/cadence-innovus-stylus-flowsetup/hard-link-to-sfk-working-directory.py** can set up the hardlinks. Note that the next time you clean or re-run the rflowgen flowsetup step, however, the hardlinks will be severed.

Abhik-Notes

`mflowgen run --update`
make clean-<N> # or by name: make clean-cadence-innovus-flowsetup

Use the templates to replace the hard coded path in setup.yamal (https://github.com/radway-lab/mflowgen-riscv-tapeout/blob/64ebeda125ef40ef134e2365c3f2677a5fb5a1ee/nodes/cadence-innovus-stylus-flowsetup/setup.yaml)

Generate setup.yamal -> construct.py (using ginga)
Possible to move in all the possible parts which can be in adk.tcl ()

1. antenna command before Signoff to fix antenna violation (add diode cells - part of eco flow) (post route)
Check if innovas have the script to fix these antenna errors.

2. Add lec step and compare the RTL and synthesis (try getting conformal script from genius it itself–)
3. Antennas fix
4. LEC step (after synthesis- check on synthesis tool)
5. Multicorner - typical typical

Flowtool – predict summary -file flow.yaml

Abhik-Mflowen Gen - FoundationFlow

Fixes for the `cadence-innovus-postroute and cadence-innovus-postroute_hold`
Problem: Scripts were not generated for postroute & postroute_hold (Cadence support: ref.)

Issue: Innovus Foundation Flow v23+ explicitly removes the postroute step from vars(steps) when using flat mode

```
# In utils.tcl (inside set_steps for "flat" mode):
if {[lindex [split $vars(version) "."] 0] > 22} {
    set vars(steps) [join [lreplace $vars(steps) ... "postroute" ...]]
}
```

This meant `gen_flow.tcl` never generated `innovus-foundation-flow/INNOVUS/run_postroute.tcl`, causing `cadence-innovus-postroute/scripts/main.tcl` to fail when it tried to `source` that file.
The `vars(postroute,restore_design,skip) true` setting in `setup.tcl` was already **correct** — `START.tcl` handles design restore via `restore-design.tcl`, so the generated script should not also restore.

Fix: Two changes
1. setup.tcl:418-422
Added an explicit vars(steps) list that includes postroute:

```
set vars(steps) {init place prects cts postcts postcts_hold route postroute postroute_hold signoff}
```

2. run.sh:45
Switched gen_flow from -m flat to -m user:

In user mode, set_steps only validates that vars(steps) exists — it does not prune or modify it
In flat mode, set_steps overrides vars(steps) and removes postroute for v23+
All other behavior is identical (user_mode is internally set to flat for both modes)

```
cd 4-cadence-genus-synthesis && if [[ -e mflowgen-check-postconditions.py ]]; then
./mflowgen-check-postconditions.py || exit 1; fi && cd ..

    > Checking postconditions for node "cadence-genus-synthesis"

pytest -q -rA --disable-warnings --tb=short --color=yes --noconftest
./mflowgen-check-postconditions.py
......F
[100%]
================================================================================
== FAILURES
================================================================================
__________________________________________________________________________________
```

```
test_6_
_____________________________________________________________________________
mflowgen-check-postconditions.py:36: in test_6_
    assert 'create_clock' in File( 'outputs/design.sdc' )
E   AssertionError: assert 'create_clock' in File( 'outputs/design.sdc' )
E    +  where File( 'outputs/design.sdc' ) = File('outputs/design.sdc')
=============================================================================
=== PASSES
=============================================================================
=== short test summary info
=============================================================================
PASSED mflowgen-check-postconditions.py::test_0_
PASSED mflowgen-check-postconditions.py::test_1_
PASSED mflowgen-check-postconditions.py::test_2_
PASSED mflowgen-check-postconditions.py::test_3_
PASSED mflowgen-check-postconditions.py::test_4_
PASSED mflowgen-check-postconditions.py::test_5_
FAILED mflowgen-check-postconditions.py::test_6_ - AssertionError: assert 'create_clock' in File(
'outputs/design.sdc' )
1 failed, 6 passed in 0.08s

make: *** [Makefile:374: 4-cadence-genus-synthesis/.postconditions.stamp] Error 1
```

Why `create_generated_clock` doesn't work for this port:

`create_generated_clock` is only valid when a **master clock already exists upstream** of the source pin. It defines a *derived* clock (e.g., after a clock divider, PLL, or across a hierarchy boundary).

`clock_uncore` is the **top-level primary clock input** — there is nothing upstream of it. Using `create_generated_clock -source [get_ports clock_uncore] ... [get_ports clock_uncore]` with source = target is degenerate and Genus silently discards it, leaving the output `design.sdc` with no clock at all.

When `create_generated_clock` IS appropriate here:

- In a parent chip where a PLL or clock mux drives `clock_uncore` — you'd use it on the PLL output pin
- In Innovus-specific SDC overrides for sub-blocks that receive a derived clock from a chip-level network

Tried creating clock with this command but it didn't worked, as clock was not created in the design.sdc

```
create_generated_clock -name ${clock_name} \
                       -source [get_ports ${clock_net}] \
```

```
                        -divide_by 1 \
                        [get_ports ${clock_net}]
```

cd 4-cadence-genus-synthesis && if [[-e mflowgen-check-postconditions.py]]; then
./mflowgen-check-postconditions.py || exit 1; fi && cd .. _ ==========

> Checking postconditions for node "cadence-genus-synthesis"
===

pytest -q -rA --disable-warnings --tb=short --color=yes --noconftest
./mflowgen-check-postconditions.py ===
.....F
[100%] ===
dc'))
==========
======== FAILURES
==========
=========
s/design.s
_____
_____ test_6_
_____=_____
_____ put
mflowgen-check-postconditions.py:36: in test_6_
= 02-Mar-26
assert 'create_clock' in File('outputs/design.sdc') or 'create_generated_clock' in File(
'outputs/design.sdc')
E AssertionError: assert ('create_clock' in File('outputs/design.sdc') or 'create_generated_clock' in
File('outputs/design.sdc'))
E + where File('outputs/design.sdc') = File('outputs/design.sdc')
out 02-Mar-26
E + and File('outputs/design.sdc') = File('outputs/design.sdc')
==========
========= PASSES
==========
-26==========
==========
= short test summary info
==========
= '
PASSED mflowgen-check-postconditions.py::test_0_
ar-
PASSED mflowgen-check-postconditions.py::test_1_
PASSED mflowgen-check-postconditions.py::test_2_
PASSED mflowgen-check-postconditions.py::test_3_
PASSED mflowgen-check-postconditions.py::test_4_
PASSED mflowgen-check-postconditions.py::test_5_
r
FAILED mflowgen-check-postconditions.py::test_6_ - AssertionError: assert ('create_clock' in File(

```
'outputs/design.sdc' ) or 'create_generated_clock' in File(
'outputs/design.sdc' ))
1 failed, 6 passed in 0.09s

make: *** [Makefile:374: 4-cadence-genus-synthesis/.postconditions.stamp] Error 1
```

Step 16:

```
(conda-mflowgen-env) [abhikkum@acgcascade52 build]$ make 16
mkdir -p 16-cadence-innovus-debug-calibre/outputs && set -o pipefail &&
/home/abhikkum/mflowgen-riscv-tapeout/mflowgen/scripts/mflowgen-letters -c -t
cadence-innovus-debug-calibre && chmod +x
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-run && cp -f
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-run 16-cadence-innovus-debug-calibre &&
if [[ -e .mflowgen/16-cadence-innovus-debug-calibre/mflowgen-debug ]]; then chmod +x
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-debug && cp -f
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-debug 16-cadence-innovus-debug-calibre;
fi && if [[ -e .mflowgen/16-cadence-innovus-debug-calibre/mflowgen-check-preconditions.py ]]; then
chmod +x .mflowgen/16-cadence-innovus-debug-calibre/mflowgen-check-preconditions.py && cp -f
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-check-preconditions.py
16-cadence-innovus-debug-calibre; fi && if [[ -e
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-check-postconditions.py ]]; then chmod +x
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-check-postconditions.py && cp -f
.mflowgen/16-cadence-innovus-debug-calibre/mflowgen-check-postconditions.py
16-cadence-innovus-debug-calibre; fi && cd 16-cadence-innovus-debug-calibre && if [[ -e
mflowgen-check-preconditions.py ]]; then ./mflowgen-check-preconditions.py || exit 1; fi &&
./mflowgen-run 2>&1 | tee mflowgen-run.log || exit 1 && cd .. && touch -c
16-cadence-innovus-debug-calibre/outputs/*


                _                 _                                  _   _   _
  ____   __ _    __| |              | |__    _   _    __ _           ____    __ _  | | (_) | |__    _ __     ____
 / __|  / _` |  / _` |              | '_ \  | | | |  / _` |  _______   / __|  / _` | | | | | | '_ \  | '__|  / _ \
| (__  | (_| | | (_| |      _   _   | |_) | | |_| | | (_| | |_______| | (__  | (_| | | | | | | | |_) | | |    | __/
 \____|  \__,_|  \__,_|     (_) (_)  |_.__/   \__,_|  \__, |          \____|  \__,_| |_| |_| |_.__/  |_|     \____|
                                                      |___/

2026/02/18 21:20:27 WARNING This OS does not appear to be a Cadence supported Linux
configuration.
2026/02/18 21:20:27 For more info, please run CheckSysConf in
<cdsRoot/tools.lnx86/bin/checkSysConf <productId>

Cadence Innovus(TM) Implementation System.
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```

```
Version:        v23.10-p003_1, built Thu Feb 1 13:54:57 PST 2024
Options:        -overwrite -init innovus-foundation-flow/custom-scripts/run-debug.tcl -log
logs/debug.log
Date:           Wed Feb 18 21:20:27 2026
Host:           acgcascade52 (x86_64 w/Linux 4.18.0-553.56.1.el8_10.x86_64)
(26cores*104cpus*Intel(R) Xeon(R) Gold 6230R CPU @ 2.10GHz 36608KB)
OS:             Rocky Linux 8.10 (Green Obsidian)

License:
                [21:20:27.060274] Configured Lic search path (23.02-s003):
1711@head.hep.upenn.edu:1717@potato.cis.upenn.edu

                invs    Innovus Implementation System   23.1    checkout succeeded
                8 CPU jobs allowed with the current license(s). Use setMultiCpuUsage to set your required
CPU count.
Create and set the environment variable TMPDIR to
/home/abhikkum/mflowgen-riscv-tapeout/build/16-cadence-innovus-debug-calibre/innovus_temp_18
10400_acgcascade52_abhikkum_hQH8oC.

Change the soft stacksize limit to 0.2%RAM (771 mbytes). Set global soft_stack_size_limit to
change the value.
Sourcing startup file ./enc.tcl
**ERROR: (IMPSYT-6689): Invalid return code while sourcing "./enc.tcl". Check and correct your Tcl
script regarding below message:
 can't read "env(CALIBRE_HOME)": no such element in array

**INFO:  MMMC transition support version v31-84

[INFO] Loading Pegasus 23.20 fill procedures
Sourcing file "innovus-foundation-flow/custom-scripts/run-debug.tcl" ...
#% Begin load design ... (date=02/18 21:20:53, mem=1781.3M)
Set Default Input Pin Transition as 0.1 ps.
Loading design 'ChipTop' saved by 'Innovus' '23.10-p003_1' on 'Wed Feb 18 15:26:39 2026'.
% Begin Load MMMC data ... (date=02/18 21:20:54, mem=1783.1M)
% End Load MMMC data ... (date=02/18 21:20:54, total cpu=0:00:00.0, real=0:00:00.0, peak
res=1783.7M, current mem=1783.7M)

Loading LEF file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/libs/lef/rtk-tech.lef ...

Loading LEF file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/libs/lef/stdcells.lef ...
Set DBUPerIGU to M2 pitch 380.

##  Check design process and node:
##  Both design process and tech node are not set.
```

```
Loading view definition file from
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/viewDefinition.tcl
Reading libs_typical timing library
'/home/abhikkum/mflowgen-riscv-tapeout/build/1-freepdk-45nm/view-standard/stdcells.lib' ...
Read 135 cells in library 'NangateOpenCellLibrary'
*** End library_loading (cpu=0.01min, real=0.02min, mem=41.0M, fe_cpu=0.31min,
fe_real=0.47min, fe_mem=1686.4M) ***
% Begin Load netlist data ... (date=02/18 21:20:55, mem=1801.1M)
*** Begin netlist parsing (mem=1686.4M) ***
Created 135 new cells from 1 timing libraries.
Reading netlist ...
Backslashed names will retain backslash and a trailing blank character.
Reading verilogBinary netlist
'/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/vbin/ChipTop.v.bin'

*** Memory Usage v#2 (Current mem = 1873.383M, initial mem = 828.582M) ***
*** End netlist parsing (cpu=0:00:01.1, real=0:00:01.0, mem=1873.4M) ***
% End Load netlist data ... (date=02/18 21:20:56, total cpu=0:00:01.1, real=0:00:01.0, peak
res=1988.6M, current mem=1983.3M)
Set top cell to ChipTop.
Hooked 135 DB cells to tlib cells.
Starting recursive module instantiation check.
No recursion found.
Building hierarchical netlist for Cell ChipTop ...
***** UseNewTieNetMode *****.
*** Netlist is unique.
** info: there are 18626 modules.
** info: there are 415848 stdCell insts.
** info: there are 0 insts with no signal pins.

*** Memory Usage v#2 (Current mem = 2216.797M, initial mem = 828.582M) ***
*info: set bottom ioPad orient R0
Start create_tracks
Generated pitch 1.68 in metal10 is different from 1.6 defined in technology file in preferred direction.
Generated pitch 0.84 in metal8 is different from 0.8 defined in technology file in preferred direction.
Generated pitch 0.84 in metal7 is different from 0.8 defined in technology file in preferred direction.
Loading preference file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/gui.pref.tcl ...
##  Process: 45            (User Set)
##     Node: (not set)

##  Check design process and node:
##  Design tech node is not set.

Applying the recommended capacitance filtering threshold values for 45nm process node:
total_c_th=0, relative_c_th=1 and coupling_c_th=0.1.
       These values will be used by all post-route extraction engines, including TQuantus, IQuantus
```

```
and Quantus QRC extraction.
      Capacitance filtering mode(-capFilterMode option of the setExtractRCMode) is 'relAndCoup' for all engines.
      The accuracy mode for postRoute effortLevel low extraction will be set to 'high'.
      Default value for EffortLevel(-effortLevel option of the setExtractRCMode) in postRoute extraction mode will be 'medium' if Quantus QRC technology file is specified else 'low'.
setAnalysisMode -usefulSkew already set.
**WARN: (IMPOPT-3602):  The specified path group name Reg2Reg is not defined.
Type 'man IMPOPT-3602' for more detail.
Effort level <high> specified for Reg2Reg path_group
Slack adjustment of -0 applied on Reg2Reg path_group
**WARN: (IMPOPT-3602):  The specified path group name Reg2Out is not defined.
Type 'man IMPOPT-3602' for more detail.
Slack adjustment of -0 applied on Reg2Out path_group
**WARN: (IMPOPT-3602):  The specified path group name Reg2ClkGate is not defined.
Type 'man IMPOPT-3602' for more detail.
Slack adjustment of -0 applied on Reg2ClkGate path_group
**WARN: (IMPOPT-3602):  The specified path group name In2Reg is not defined.
Type 'man IMPOPT-3602' for more detail.
Slack adjustment of -0 applied on In2Reg path_group
**WARN: (IMPOPT-3602):  The specified path group name In2Out is not defined.
Type 'man IMPOPT-3602' for more detail.
Slack adjustment of -0 applied on In2Out path_group
#WARNING (NRIF-95) Option setNanoRouteMode -routeTopRoutingLayer has invalid value "0". Reset to max routing layer "10".
#WARNING (NRIF-91) Option setNanoRouteMode -route_top_routing_layer is obsolete. It will continue to work for the current release. To ensure compatibility with future releases, use option setDesignMode -topRoutingLayer instead.

viaInitial starts at Wed Feb 18 21:20:57 2026
viaInitial ends at Wed Feb 18 21:20:57 2026
Setting -ignore_DRC to 0. ViaGen will do DRC check while creating vias.
Setting -viarule_preference to default order. ViaGen will consider via rule priority based on the order of appearance in the technology file.
AAE_INFO: switching setDelayCal -siAware from false to true ...
AAE_INFO: The setting is changed from false to true in setDelayCalMode -SIAware
[21:20:57.434011] Periodic Lic check successful
[21:20:57.958321] Feature usage summary:
[21:20:57.958322] Innovus_Impl_System
[21:20:57.958322] Tempus_Timing_Signoff_XL

This command "restoreDesign /home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.checkpoint/save.enc.dat ..." required an extra checkout of license tpsxl.
Additional license(s) checked out: 1 'Tempus_Timing_Signoff_XL' license(s)
**WARN: (IMPPTN-867):   Found use of setPinAssignMode -maxLayer. This will continue to work in this release; however, it is recommended to use setDesignMode -topRoutingLayer instead.
**WARN: (IMPPSP-1003):  Found use of 'setRouteMode -earlyGlobalMaxRouteLayer'. This will continue to work in this release; however, it is recommended to use 'setDesignMode -topRoutingLayer' instead.
```

```
Change floorplan default-technical-site to 'FreePDK45_38x28_10R_NP_162NW_34O'.
*Info: initialize multi-corner CTS.
Total number of combinational cells: 93
Total number of sequential cells: 29
Total number of tristate cells: 6
Total number of level shifter cells: 0
Total number of power gating cells: 0
Total number of isolation cells: 0
Total number of power switch cells: 0
Total number of pulse generator cells: 0
Total number of always on buffers: 0
Total number of retention cells: 0
Total number of physical cells: 7
List of usable buffers: BUF_X1 BUF_X2 BUF_X4 BUF_X8 BUF_X16 BUF_X32 CLKBUF_X1
CLKBUF_X2 CLKBUF_X3
Total number of usable buffers: 9
List of unusable buffers:
Total number of unusable buffers: 0
List of usable inverters: INV_X1 INV_X2 INV_X4 INV_X8 INV_X16 INV_X32
Total number of usable inverters: 6
List of unusable inverters:
Total number of unusable inverters: 0
List of identified usable delay cells:
Total number of identified usable delay cells: 0
List of identified unusable delay cells:
Total number of identified unusable delay cells: 0
No delay cells were detected in the set of buffers. Buffers will be used to fix hold violations.
Reading floorplan file -
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/ChipTop.fp.gz (mem = 2483.8M).
% Begin Load floorplan data ... (date=02/18 21:20:58, mem=2602.6M)
2026/02/18 21:20:58 WARNING This OS does not appear to be a Cadence supported Linux
configuration.
2026/02/18 21:20:58 For more info, please run CheckSysConf in
<cdsRoot/tools.lnx86/bin/checkSysConf <productId>
2026/02/18 21:20:58 WARNING This OS does not appear to be a Cadence supported Linux
configuration.
2026/02/18 21:20:58 For more info, please run CheckSysConf in
<cdsRoot/tools.lnx86/bin/checkSysConf <productId>
*info: reset 631755 existing net BottomPreferredLayer and AvoidDetour
Deleting old partition specification.
Set FPlanBox to (0 0 2551700 2549120)
Start create_tracks
Generated pitch 1.68 in metal10 is different from 1.6 defined in technology file in preferred direction.
Generated pitch 0.84 in metal8 is different from 0.8 defined in technology file in preferred direction.
Generated pitch 0.84 in metal7 is different from 0.8 defined in technology file in preferred direction.
 ... processed partition successfully.
Reading binary special route file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/ChipTop.fp.spr.gz (Created by Innovus v23.10-p003_1 on Wed Feb 18 15:26:33
```

```
2026, version: 1)
Ending "Read special route file" (total cpu=0:00:00.0, real=0:00:01.0, peak res=2607.9M, current
mem=2607.9M)
There are 22288 nets with weight being set
There are 22297 nets with bottomPreferredRoutingLayer being set
There are 22288 nets with avoidDetour being set
Extracting standard cell pins and blockage ......
Pin and blockage extraction finished
Delete all existing relative floorplan constraints.
% End Load floorplan data ... (date=02/18 21:20:59, total cpu=0:00:00.5, real=0:00:01.0, peak
res=2608.7M, current mem=2608.7M)
Reading congestion map file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/ChipTop.route.congmap.gz ...
% Begin Load SymbolTable ... (date=02/18 21:20:59, mem=2608.7M)
2026/02/18 21:20:59 WARNING This OS does not appear to be a Cadence supported Linux
configuration.
2026/02/18 21:20:59 For more info, please run CheckSysConf in
<cdsRoot/tools.lnx86/bin/checkSysConf <productId>
Suppress "**WARN ..." messages.
2026/02/18 21:20:59 WARNING This OS does not appear to be a Cadence supported Linux
configuration.
2026/02/18 21:20:59 For more info, please run CheckSysConf in
<cdsRoot/tools.lnx86/bin/checkSysConf <productId>
Un-suppress "**WARN ..." messages.
% End Load SymbolTable ... (date=02/18 21:21:00, total cpu=0:00:01.6, real=0:00:01.0, peak
res=2717.1M, current mem=2717.1M)
Loading place ...
% Begin Load placement data ... (date=02/18 21:21:00, mem=2717.1M)
Reading placement file -
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/ChipTop.place.gz.
** Reading stdCellPlacement_binary (Created by Innovus v23.10-p003_1 on Wed Feb 18 15:26:33
2026, version# 2) ...
Read Views for adaptive view pruning ...
Read 0 views from Binary DB for adaptive view pruning
*** applyConnectGlobalNets disabled.
*** Completed restorePlace (cpu=0:00:02.2 real=0:00:03.0 mem=2793.1M) ***
Total net length = 7.026e+06 (3.539e+06 3.487e+06) (ext = 4.012e+04)
% End Load placement data ... (date=02/18 21:21:03, total cpu=0:00:02.3, real=0:00:03.0, peak
res=2911.6M, current mem=2908.5M)
2026/02/18 21:21:03 WARNING This OS does not appear to be a Cadence supported Linux
configuration.
2026/02/18 21:21:03 For more info, please run CheckSysConf in
<cdsRoot/tools.lnx86/bin/checkSysConf <productId>
Reading PG file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/ChipTop.pg.gz, version#2, (Created by Innovus v23.10-p003_1 on       Wed Feb
18 15:26:33 2026)
*** Completed restorePGFile (cpu=0:00:00.3 real=0:00:00.0 mem=2806.1M) ***
```

```
% Begin Load routing data ... (date=02/18 21:21:03, mem=2926.2M)
Reading routing file -
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.checkpoint/save.enc.dat/ChipTop.route.gz.
2026/02/18 21:21:03 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/02/18 21:21:03 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>
Reading Innovus routing data (Created by Innovus v23.10-p003_1 on Wed Feb 18 15:26:33 2026 Format: 23.1) ...
*** Total 593123 nets are successfully restored.
*** Completed restoreRoute (cpu=0:00:03.4 real=0:00:04.0 mem=3407.1M) ***
% End Load routing data ... (date=02/18 21:21:07, total cpu=0:00:03.5, real=0:00:04.0, peak res=3536.8M, current mem=3536.8M)
TAT_INFO: ::restoreCongMap REAL = 0 : CPU = 0 : MEM = 0.
Reading property file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.checkpoint/save.enc.dat/ChipTop.prop
*** Completed restoreProperty (cpu=0:00:00.2 real=0:00:00.0 mem=3413.1M) ***
2026/02/18 21:21:07 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/02/18 21:21:07 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>
2026/02/18 21:21:07 WARNING This OS does not appear to be a Cadence supported Linux configuration.
2026/02/18 21:21:07 For more info, please run CheckSysConf in <cdsRoot/tools.lnx86/bin/checkSysConf <productId>
Reading dirtyarea snapshot file
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.checkpoint/save.enc.dat/ChipTop.db.da.gz (Create by Innovus v23.10-p003_1 on Wed Feb 18 15:26:37 2026, version: 7).
Set Default Input Pin Transition as 0.1 ps.
Extraction setup Started for TopCell ChipTop
Initializing multi-corner RC extraction with 1 active RC Corners ...
Reading Capacitance Table File
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.checkpoint/save.enc.dat/libs/mmmc/rtk-typical.captable ...
Cap table was created using Encounter 08.10-p004_1.
Process name: master_techFreePDK45.
Importing multi-corner RC tables ...
Summary of Active RC-Corners :

 Analysis View: analysis_default
    RC-Corner Name        : typical
    RC-Corner Index       : 0
    RC-Corner Temperature : 25 Celsius
    RC-Corner Cap Table   :
'/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.checkpoint/save.enc.dat/libs/mmmc/rtk-typical.captable'
    RC-Corner PreRoute Res Factor         : 1
```

```
    RC-Corner PreRoute Cap Factor         : 1
    RC-Corner PostRoute Res Factor        : 1 {1 1 1}
    RC-Corner PostRoute Cap Factor        : 1 {1 1 1}
    RC-Corner PostRoute XCap Factor       : 1 {1 1 1}
    RC-Corner PreRoute Clock Res Factor   : 1   [Derived from postRoute_res (effortLevel low)]
    RC-Corner PreRoute Clock Cap Factor   : 1   [Derived from postRoute_cap (effortLevel low)]
    RC-Corner PostRoute Clock Cap Factor  : 1 {1 1 1}   [Derived from postRoute_cap (effortLevel
low)]
    RC-Corner PostRoute Clock Res Factor  : 1 {1 1 1}   [Derived from postRoute_res (effortLevel
low)]
    RC-Corner PostRoute Clock coupling capacitance Factor : 1 {1 1 1}
Initializing multi-corner capacitance tables ...
Initializing multi-corner resistance tables ...
Start generating vias ..
#Skip building auto via since it is not turned on.
Extracting standard cell pins and blockage ......
Pin and blockage extraction finished
Via generation completed.
% Begin Load power constraints ... (date=02/18 21:21:12, mem=3704.6M)
source
/home/abhikkum/mflowgen-riscv-tapeout/build/12-cadence-innovus-signoff/checkpoints/design.chec
kpoint/save.enc.dat/ChipTop_power_constraints.tcl
'set_default_switching_activity' finished successfully.
** WARN:  (VOLTUS_POWR-3212): The 'set_power_analysis_mode -leakage_power_view
|-dynamic_power_view|-analysis_view' will be obsolete in 18.20 release. Use 'set_analysis_view
-leakage <> | -dynamic <>' to set leakage and dynamic power views.

% End Load power constraints ... (date=02/18 21:21:12, total cpu=0:00:00.0, real=0:00:00.0, peak
res=3711.7M, current mem=3711.7M)
% Begin load AAE data ... (date=02/18 21:21:14, mem=3829.2M)
**WARN: (IMPESI-3505):  setDelayCalMode -eng_enablePrePlacedFlow false (default=false) will be
obsoleted along with its sgs2set equivalent.  This parameter will continue to be supported in the
current release, but will be removed in the next major release of the software.
AAE DB initialization (MEM=3830.25 CPU=0:00:00.1 REAL=0:00:00.0)
% End load AAE data ... (date=02/18 21:21:14, total cpu=0:00:00.6, real=0:00:00.0, peak
res=3889.7M, current mem=3889.7M)
Restoring CCOpt config...
  Extracting original clock gating for chip_clock_uncore...
    clock_tree chip_clock_uncore contains 175203 sinks and 17922 clock gates.
  Extracting original clock gating for chip_clock_uncore done.
  The skew group chip_clock_uncore/constraints_default was created. It contains 175203 sinks and
1 sources.
  The skew group chip_clock_uncore/constraints_default was created. It contains 175203 sinks and
1 sources.
Restoring CCOpt config done.
Total number of combinational cells: 93
Total number of sequential cells: 29
Total number of tristate cells: 6
Total number of level shifter cells: 0
Total number of power gating cells: 0
```

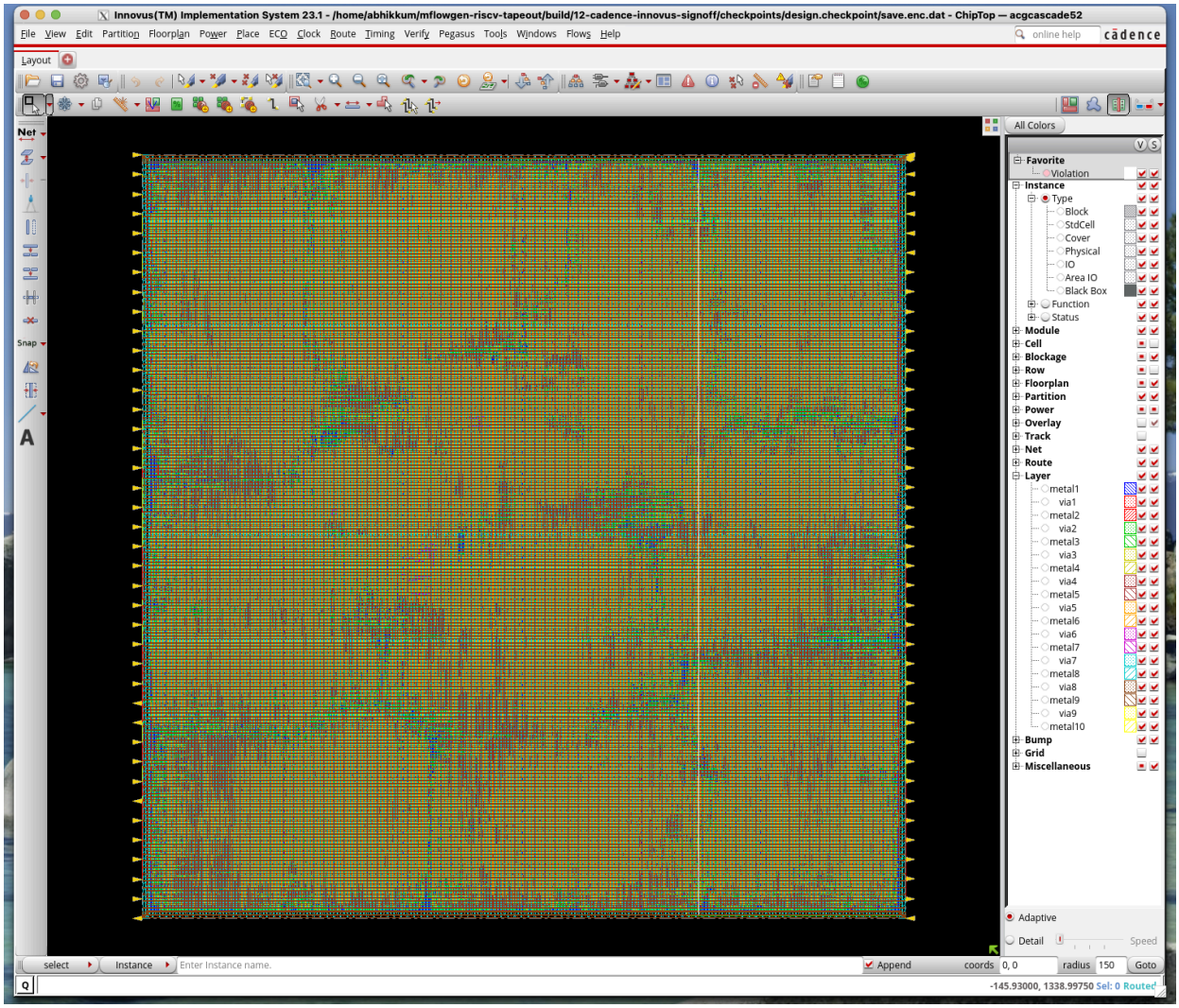
```
Total number of isolation cells: 0
Total number of power switch cells: 0
Total number of pulse generator cells: 0
Total number of always on buffers: 0
Total number of retention cells: 0
Total number of physical cells: 7
List of usable buffers: BUF_X1 BUF_X2 BUF_X4 BUF_X8 BUF_X16 BUF_X32 CLKBUF_X1
CLKBUF_X2 CLKBUF_X3
Total number of usable buffers: 9
List of unusable buffers:
Total number of unusable buffers: 0
List of usable inverters: INV_X1 INV_X2 INV_X4 INV_X8 INV_X16 INV_X32
Total number of usable inverters: 6
List of unusable inverters:
Total number of unusable inverters: 0
List of identified usable delay cells:
Total number of identified usable delay cells: 0
List of identified unusable delay cells:
Total number of identified unusable delay cells: 0
No delay cells were detected in the set of buffers. Buffers will be used to fix hold violations.
**WARN: (IMPCTE-107):   The following globals have been obsoleted since version . They will be
removed in the next release.
timing_aocv_enable_gba_combine_launch_capture
timing_enable_backward_compatible_latch_thru_mt_mode
timing_enable_separate_device_slew_effect_sensitivities
#% End load design ... (date=02/18 21:21:20, total cpu=0:00:27.5, real=0:00:27.0, peak
res=4503.5M, current mem=4278.1M)

*** Summary of all messages that are not suppressed in this session:
Severity  ID               Count  Summary
WARNING   IMPPTN-867           1  Found use of %s. This will continue to w...
WARNING   IMPESI-3505          1  setDelayCalMode -eng_%s (default=%s) wil...
WARNING   IMPOPT-3602          5  The specified path group name %s is not ...
WARNING   NRIF-91              1  Option setNanoRouteMode -route_top_routi...
WARNING   NRIF-95              1  Option setNanoRouteMode -routeTopRouting...
WARNING   IMPCTE-107           1  The following globals have been obsolete...
WARNING   IMPPSP-1003          1  Found use of '%s'. This will continue to...
*** Message Summary: 11 warning(s), 0 error(s)

The timeout for a remote job to respond is 3600 seconds.
Submit command for task runs will be: local
<FF> Plugin -> always_source_tcl
<FF> LOADING 'always_source_tcl' PLUG-IN FILE(s)
<FF> -> innovus-foundation-flow/custom-scripts/always-source.tcl
innovus 1>close
touch 16-cadence-innovus-debug-calibre/.execstamp.execute-phony
cd 16-cadence-innovus-debug-calibre && if [[ -e mflowgen-check-postconditions.py ]]; then
./mflowgen-check-postconditions.py || exit 1; fi && cd ..
touch 16-cadence-innovus-debug-calibre/.postconditions.stamp
```

Issue with caliber side data format different may be

With x11 forwarding



Shreyaans - Notes

03/04
Nano route - logic values
SDC - set clock
UPF - set ground power rails

Create clock or generate clock - referenced to the same clock → for synch clock in different modules
Get a master clock form a pin (if offchip)→ connect it to generated clock like clock_uncore

Fixing post route steps in the innovus foundation flow. Clean all and re-run.

Sign off - timing rerun

U can sub-execute tempus from innovus. Does STA.

Voltus → EMIR analysis electron migration and IR drop. Determines what voltage every single cell sees.

Rerun voltus for IR drop across all cells and then Rerun Tempus with the voltus annotation to check timing with IR drop.

Hierarchical design - closing timings for hierarchical designs. Innovus + Stylus support hierarchical retiming. Not supported by Mflowgen and Innovus foundation flow.

*fix - mflowgen assumes the standard cell library is just 1 file.

Adk plugin fix to create a symlink and not hard copy

For stylus flow → floorplanning and PNR run successfully then need to debug CTS.

ADK file changes for stylus :

Issue: Static `setup.yaml` incompatible with ADK-agnostic flow

The Stylus FlowKit expects library files, RC corners, and cell names to be declared in `setup.yaml`. The existing `run.sh` simply symlinks a hardcoded `setup.yaml` that references `stdcells-bc.lib` and `stdcells-wc.lib` (BC/WC timing corners) and a `qrc_tech_file` — none of which exist in freepdk-45nm. Cell names like `FILLCELL_X32` and `ANTENNA_X1` are also hardcoded, meaning the file breaks entirely when switching to a different ADK like TSMC 40nm.

The foundation flow avoided this by using TCL `glob` patterns in `setup.tcl` to discover whatever libs exist and conditionally build the MMMC setup — only adding BC/WC corners if those files were actually present.

Fix: Dynamic `setup.yaml` generation

Replaced the static `ln -sf ../setup.yaml` in `run.sh` with a call to `generate-setup-yaml.py` which:

- Globs `inputs/adk/` for TT/BC/WC `.lib` files — only includes corners that actually exist
- Checks for `qrc_tech_file` or `captable` — includes only if present
- Parses `inputs/adk/adk.tcl` to extract cell names (`ADK_FILLER_CELLS`, `ADK_TIE_CELLS`, `ADK_ANTENNA_CELL`, `ADK_PROCESS`, `ADK_MAX_ROUTING_LAYER_INNOVUS`) instead of hardcoding them
- Writes a correct `setup.yaml` at runtime

Shreyaans - Foundation Flow Postroute fix

Fix cadence-innovus-postroute for Innovus 23.10

Problem:
The Innovus Foundation Flow generator (gen_flow.tcl) in Innovus 23.10 no longer generates run_postroute.tcl. Running gen_flow.tcl with 'all' only produces: init, place, cts, postcts_hold, route, signoff, debug, lec. The 'postroute' and 'postroute_hold' steps are not recognized as valid Foundation Flow steps in this version, causing step 12 (cadence-innovus-postroute) to fail with:
"ERROR: The file 'innovus-foundation-flow/INNOVUS/run_postroute.tcl' could not be located"

Fix:
1. nodes/cadence-innovus-flowsetup/run.sh
   - Changed gen_flow.tcl argument from 'all' to explicit step list: init place cts postcts_hold route signoff
   - Avoids errors from passing unrecognized step names to gen_flow.tcl

2. nodes/cadence-innovus-postroute/scripts/run_postroute.tcl (new file)
   - Custom postroute script modeled after the generated run_route.tcl
   - Runs optDesign -postRoute for setup timing closure
   - Runs optDesign -postRoute -hold for hold timing closure
   - Uses vars(postroute_extraction_effort) from vars.tcl for RC extraction

3. nodes/cadence-innovus-postroute/scripts/main.tcl
   - Updated to source scripts/run_postroute.tcl directly instead of the missing innovus-foundation-flow/INNOVUS/run_postroute.tcl

Tested on: Innovus 23.10-p003_1, freepdk-45nm, TinyRocketConfig

DRC Summary — TinyRocket, freepdk-45nm, March 18 2026

Total violations: **124**

Rule	Count	Description
Antenna.metal3	1	Antenna violation on metal3
Antenna.metal4	6	Antenna violation on metal4
Antenna.metal5	22	Antenna violation on metal5
Antenna.metal6	31	Antenna violation on metal6

Metal8.7	63	Metal8 spacing/density rule
Metal9.6	1	Metal9 spacing/density rule

Rayhan's Notes

Attempt at IR Drop Analysis implementation:

Partial implementation and rail analysis config (and some other Voltus stuff) already present in SFK (design_config.tcl under cadence-innovus-stylus-flowsetup).

Needs power grid libraries (created with set_pg_library_mode and generate_pg_library), which in turn require QRC tech files which are missing in freePDK. Might be more doable with TSMC

TSMC Integration notes:

Location of liberty files:
TSMC/HOME/N40_STD_CELLS/TSMCHOME/digital/Front_End/timing_power_noise/

Location of LEF files:
TSMC/HOME/N40_STD_CELLS/TSMCHOME/digital/Back_End/lef/

Location of documentation listing all available standard cells:
/pdk/TSMC/HOME/N40_STD_CELLS/TSMCHOME/digital/Documentation/documents/tcbn40lpbwp_200a/DB_TCBN40LPBWP_TC.pdf